

Python Basics: Operators

This notebook covers:

1. Arithmetic Operators
2. Relational (Comparison) Operators
3. Logical Operators

Note: Only Python basics are used. No conditions, loops, or advanced topics.

1. Arithmetic Operators

Arithmetic operators are used to perform mathematical calculations.

Variables for Arithmetic Operations

In [1]:

```
num1 = 20  
num2 = 5
```

```
print("num1 =", num1)  
print("num2 =", num2)
```

Addition (+)

In [2]:

```
print("Addition:", num1 + num2)
```

Subtraction (-)

In [3]:

```
print("Subtraction:", num1 - num2)
```

Multiplication ()*

In [4]:

```
print("Multiplication:", num1 * num2)
```

Division (/)

Always gives a decimal result.

In [5]:

```
print("Division:", num1 / num2)
```

Floor Division (//)

Gives the whole number part only.

In [6]:

```
print("Floor Division:", num1 // num2)
```

Modulus (%)

Gives the remainder.

In [7]:

```
print("Remainder:", num1 % num2)
```

Exponent ()

Used for power calculations.

In [8]:

```
print("Power:", num2 ** 2)
```

Arithmetic Operators with User Input

In [9]:

```
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))
```

```
print("Sum:", a + b)  
print("Difference:", a - b)  
print("Product:", a * b)
```

2. Relational (Comparison) Operators

Relational operators compare two values and return **True** or **False**.

Variables for Comparison

In [10]:

```
math_marks = 75  
science_marks = 65
```

```
print("Math Marks:", math_marks)  
print("Science Marks:", science_marks)
```

Equal To (==)

In [11]:

```
print(math_marks == science_marks)
```

Not Equal To (!=)

In [12]:

```
print(math_marks != science_marks)
```

Greater Than (>)

In [13]:

```
print(math_marks > science_marks)
```

Less Than (<)

In [14]:

```
print(math_marks < science_marks)
```

Greater Than or Equal To (>=)

In [15]:

```
print(math_marks >= 75)
```

Less Than or Equal To (<=)

In [16]:

```
print(science_marks <= 70)
```

3. Logical Operators

Logical operators are used to combine multiple conditions.

- and
- or
- not

Boolean Variables

In [17]:

```
has_id_card = True  
has_uniform = False
```

```
print(has_id_card)  
print(has_uniform)
```

Logical AND (and)

True only if both values are True.

In [18]:

```
print(has_id_card and has_uniform)
```

Logical OR (or)

True if at least one value is True.

In [19]:

```
print(has_id_card or has_uniform)
```

Logical NOT (not)

Reverses the result.

In [20]:

```
print(not has_uniform)
```

Logical Operators with Comparisons

In [21]:

```
english_marks = 70
```

```
attendance = 80
```

```
print(english_marks >= 60 and attendance >= 75)
```

```
print(english_marks >= 60 or attendance >= 90)
```

Summary

- Arithmetic operators perform calculations
- Relational operators compare values
- Logical operators combine conditions
- All operators return results that can be printed